

JC/1 Binary HTTP Protocol

A deliberately small, persistent socket protocol for the calculator and file server assignment.

1. Connection and framing

JC/1 carries an ordered request/response stream over one TCP connection. The client opens one connection, sends one or more requests, reads one response for each, and closes after the last response. The server keeps the connection open until the peer closes it, then accepts another connection. All integers are unsigned big-endian.

Offset	Width	Field
0	2	ASCII magic JC
2	1	Version 1
3	1	Type: 1 request, 2 response
4	1	Flags, currently 0
5	1	Encoded header count
6-7	2	Reserved, must be zero
8-11	4	Payload length

The payload length counts bytes after the 12-byte header. A receiver consumes exactly that many bytes before reading the next frame. An unknown type is skipped by consuming its payload. Bad magic, version, reserved fields, or a truncated payload are malformed; a complete malformed request receives status 400.

2. Header block

Each header is a one-byte name index followed by a two-byte value length and value bytes. Index 0 is followed by a one-byte literal name length and name bytes. This provides indexed names and length-prefixed values without a dynamic compression table.

Index	Name	Index	Name
1	host	6	connection
2	user-agent	7	server
3	accept	8	date
4	content-length	9	cache-control
5	content-type	10	transfer-encoding

A request payload is: method byte (GET is 1), two-byte path length, path bytes, encoded headers, four-byte body length, and body bytes. A response payload is: two-byte status, encoded headers, four-byte body length, and body bytes. Requests require a non-empty host header and an empty body.

3. Application behavior

GET /add, /sub, /mul, and /div require exactly one integer a and one integer b. A successful response is 200 with the decimal result and no trailing newline. Division by zero, missing or repeated parameters, invalid integers, missing host, and malformed requests are 400. Unknown routes such as /pow are 404; unsupported methods such as POST are 405.

All other GET paths are resolved beneath the configured document root. An existing regular file returns 200 with its exact bytes; a missing file returns 404. Paths containing .. or escaping the root are 400. Responses include content-length, content-type, connection: keep-alive, and server headers.

4. Commands

Server:

```
./serve
```

Client:

```
./b curl [-v] host:port/path [additional paths...]
```

The curl token is the required client subcommand. The client writes response bodies to stdout, writes complete request and response frame hexdumps to stderr with -v, and exits non-zero for 4xx and 5xx responses. Multiple paths use the same connection.

5. Worked byte example

For GET /add?a=2&b;=3, the first twelve request bytes are:

```
4a 43 01 01 00 04 00 00 00 00 00 3d
```

This means magic JC, version 1, request type 1, no flags, four headers, two zero reserved bytes, and a 61-byte payload. The response starts:

```
4a 43 01 02 00 04 00 00 00 00 00 2c 00 c8
```

It is a response type 2 with four headers, a 44-byte payload, and status 200 (0x00c8). The final response body length is 1 and the body byte is ASCII 5.

6. Interoperability rules

The server reads one complete frame at a time and stays on the line after each response. The client never opens a second connection during one invocation. Both sides use only the fields and static names defined here. A receiver that encounters an unknown frame type consumes its declared payload and continues with the next frame, leaving room for a future protocol version.